Maritime Safety Multi-Agent Simulation

Human-Factor Collision Avoidance & SAR
Baltic–North Sea Case Study

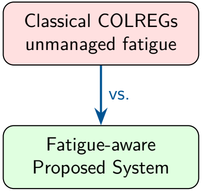
Paweł Pozorski, Michał Pytel

Warsaw University of Technology

September 6, 2026

Problem Statement

- 75–96% of maritime casualties involve **human error**
- COLREGs assume crews hear the warning and agree who gives way
- That breaks under two coupled stressors:
  - **Weather / radio:** storms cut VHF/AIS reliability
  - **Crew fatigue:** tired watchkeepers miss or delay CPA warnings
- Many ABMs treat fatigue as a side model — not linked to whether the manoeuvre happens



Research Question & Hypotheses

RQ: Does linking fatigue management to warning reliability (P_{comms}) improve coordination and safety vs classical, weather-unaware practice?

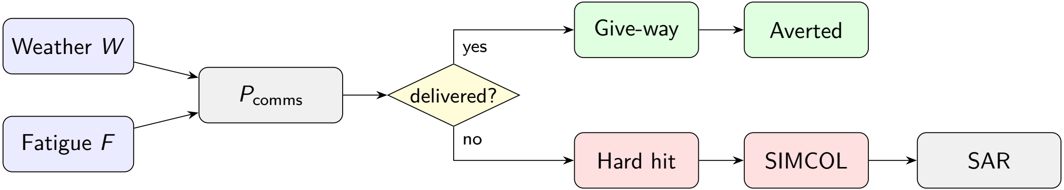
Three methods (fatigue rate m, weather slowdown, preparedness weight α_M):

- **A** Classical — $m = 1.00$, no weather slowdown, $\alpha_M = 2.0$
- **B** Shore Rest — $m = 0.82$, weather slowdown, $\alpha_M = 1.3$
- **P** Proposed — $m = 0.65$ (smart watch), same slowdown as B, $\alpha_M = 1.0$

Three primary hypotheses:

- **H1** — Blind Shore: P cuts fatal rate vs A by $\geq$20%
- **H2** — P raises P_{prep} vs A in *every* scenario
- **H3** — Blind Shore: P cuts collision rate vs A by $\geq$10%

Causal Safety Pipeline



$$P_{\text{comms}} = \min(q_a, q_b)\ \min\left(g(F_a),\ g(F_b)\right)$$

Scenarios & Scale

Scenario	N	Horizon	Stress
Calm Passage	750	14 d	Fair weather; calibration
Storm Corridor	900	≈4 d	Moving storm, comms 0.08
Blind Shore	900	7 d	Patchy shore radio 0.55
Deep Water Rescue	500	7 d	Winter SAR, slower assets

- Factorial: 4 scenarios $\times$ 3 methods $\times$ 30 seeds $=$ **360 runs**
- Fleets target concurrent *underway* AIS scale (Baltic ≈774 moving)
- Stats: Mann–Whitney U, Holm over six primary contrasts, Cliff $|\delta| \geq 0.2$

Primary KPIs

- `fatal_per_1k_hrs` — unrecovered MOB (H1)
- `collision_per_1k_hrs` — rising-edge hits (H3)
- `mean_p_prep` — crew preparedness (H2)

$$P_{\text{prep}} = \max(0, 1 - \alpha_M W)\ \max(0, 1 - 0.70\, F)$$

Also logged

- Evac activation rate
- Survival ratio, mean TTA
- IWRAP N_c stamp (geometry check)

Fatality = unrecovered person-in-water inside τ_{mob}; liferaft Lost is separate.

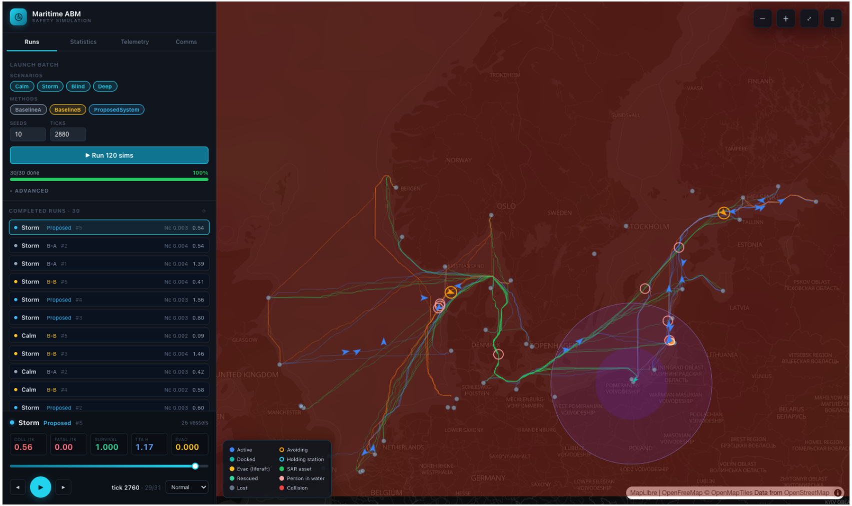
Does the simulated world look right?

Baseline A only — if this is nonsense, H1–H3 are uninterpretable.

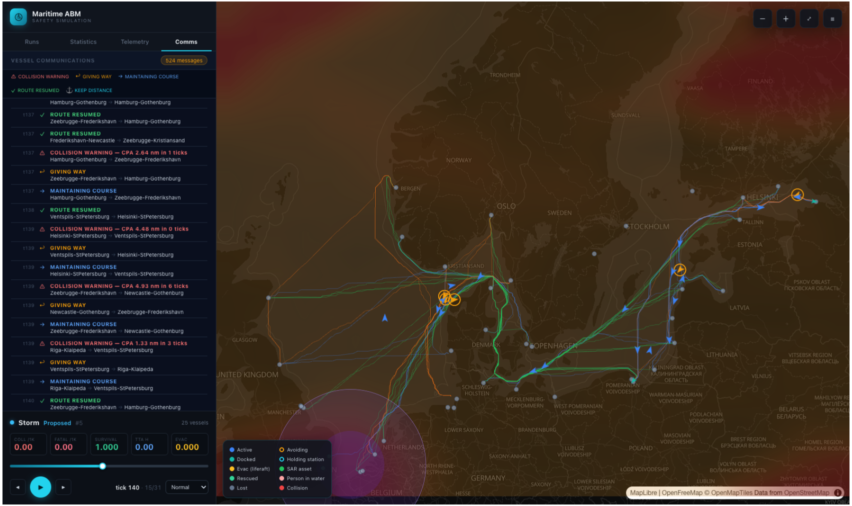
- **Calm is not a warzone:** 0.55 collisions / 1k ship-hrs, inside the 0.5–2.0 band we targeted
- **Storm actually hurts:** 3.71$\times$ Calm collisions — weather is a visible stressor
- **Blind carries the fatal mass:** 0.49 fatals vs Calm 0.10 — that is why H1/H3 live here, not in Storm

Storm is a shared radio blackout, so methods cannot separate on fatals there.

Workbench — Map & Playback



Workbench — Comms Log



Agent Platform & Software

Stack

- Engine: **Rust + krABMaga 0.6**
- Batches: Axum 0.8, Rayon, SSE
- Store: SQLite + JSONL tick logs
- UI: React 19, Vite, deck.gl 9, MapLibre
- Routes: `aisprocess` A* + Python fetchers

Agent computing

- Agent trait: `VesselAgent`, `PostTickAgent`
- Schedule: vessels order 200, post-tick 1000
- SAR / MOB / PIW: stepped *inside* post-tick (not scheduled)
- No FIPA/ACL — typed in-sim `MsgKind` log
- COLREGs dialogue, `PostTick`-mediated, lossy P_{comms}

Project Structure

Engine (`crates/simulation/`)

- `vessel.rs` — navigation, fatigue, FSM
- `state.rs` — tick / post_tick pipeline
- `comms.rs` — CPA messages & waypoints
- `simcol.rs` — collision damage
- `sar.rs` — liferaft & MOB search
- `weather.rs` — 50×50 hazard field

Repo

- `crates/api/` — Axum batches, SSE
- `frontend/` — React / deck.gl workbench
- `data/` — AIS routes, ports, coastline
- `report/` — paper + figures

Agent Types

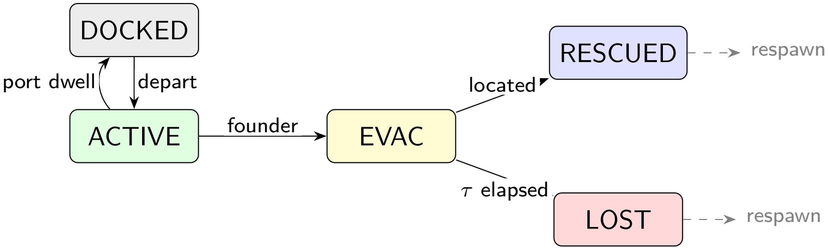
Agent	Role	Key behaviour
VesselAgent	Navigating ship	AIS route / avoidance WP; fatigue F; states Active$\to$Docked / Evac
PostTickAgent	Shared physics	Weather, CPA, following, SIMCOL, KPI, fleet top-up
RescueAgent	Liferaft SAR	Helo/Patrol: mobilise$\to$transit$\to$search$\to$embark
MobSearchAgent	Person-in-water	Same phases without embark; per-person Koopman draws
MobPerson	Casualty datum	Drift + τ_{mob} survival window

Macro-Tick Sequence

$\Delta t = 5\,\text{min}$: step 1 `VesselAgent`, then shared `post_tick` (2–13).

#	Phase	What happens
1	Navigate	AIS route or avoidance waypoint
2	Weather	Update hazard field W, storm cells
3	Grounding	Snap off the land mask
4	Fatigue / speed	Tick F; weather slowdown (B/P)
5	Comms log	`ResumeRoute` when WP queue empties
6	Avoidance	CPA pair; P_{comms} injects starboard WP
7	Following	Same-destination keep-distance
8	Collision	SIMCOL: founder $\rightarrow$ Evac, else MOB
9	SAR	Helo/patrol: search $\rightarrow$ embark / Lost
10	MOB	PIW search; fatality if τ_{mob} elapses
11	Reap	Drop Rescued / Lost
12	KPI / fleet	Record KPIs; spawn until fleet $= N$
13	Housekeeping	UTC, storm track, JSONL snapshot

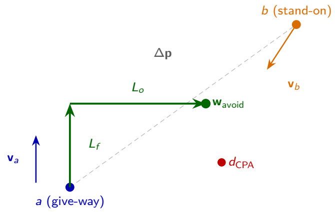
Vessel State Machine



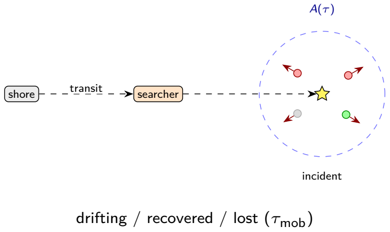
Foundering (SIMCOL) enters Evac; afloat overboard spawns MOB persons instead.

Give-Way, MOB & SAR

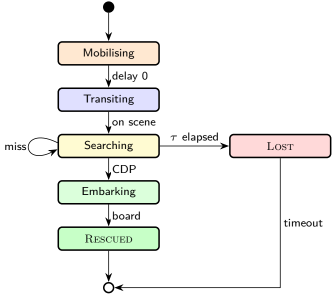
Give-way waypoint



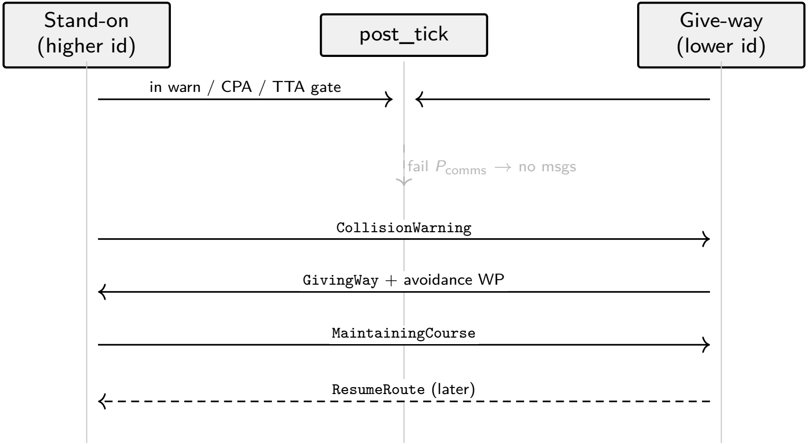
MOB search



RescuePhase



Inter-Agent Communications (CPA)



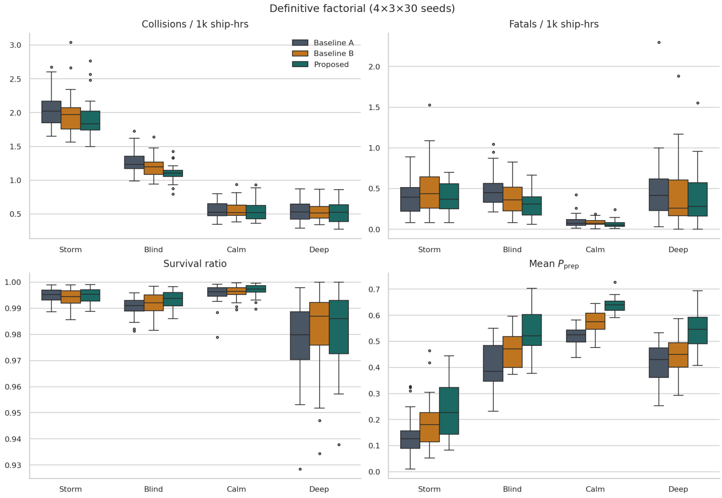
$P_{\text{comms}} = \min(q_a, q_b)\ \min(g(F_a), g(F_b))$; also `KeepDistance` on shared port approaches.

Hypothesis Outcomes

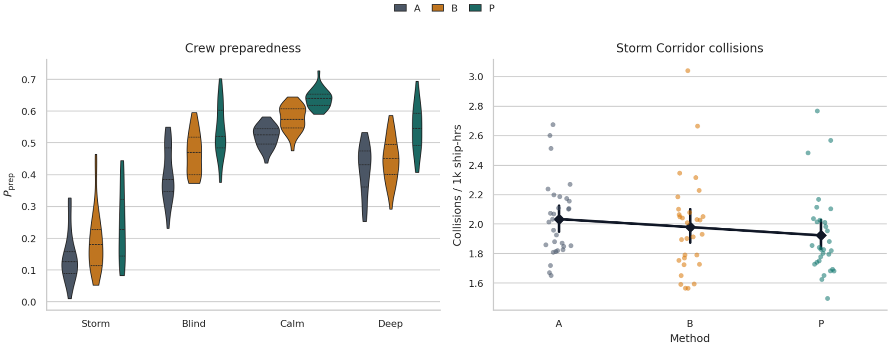
Hyp.	Contrast	Means (A→P)	δ
H1	Blind fatals	$0.486 \to 0.304$ ($\approx$37%)	$+0.53$
H2	Calm P_{prep}	$0.522 \to 0.639$	-1.00
H2	Storm P_{prep}	$0.140 \to 0.242$	-0.52
H2	Blind P_{prep}	$0.401 \to 0.536$	-0.70
H2	Deep P_{prep}	$0.415 \to 0.543$	-0.77
H3	Blind collisions	$1.26 \to 1.11$ ($\approx$12%)	$+0.54$

All six primary contrasts **confirmed** after Holm ($p_{\mathrm{corr}} < 0.05$, $|\delta| \geq 0.2$).

KPI Distributions

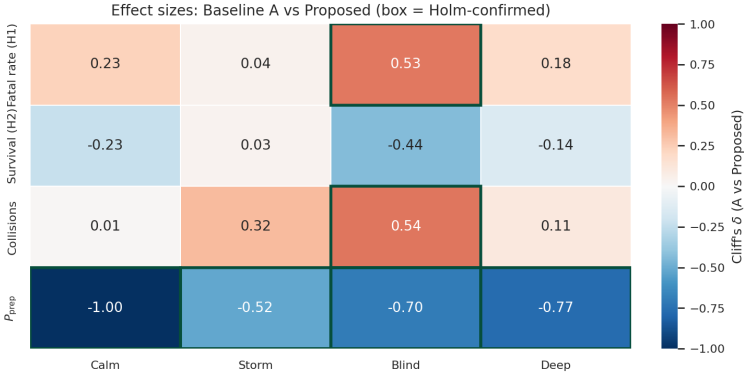


Preparedness & Storm Collisions



P_{prep} rises A<B<P everywhere (H2). Storm collisions fall only modestly — shared radio blackout.

Hypothesis Heatmap



Cliff's δ for A vs P (exploratory full battery); primary H1–H3 use the six-contrast Holm set.

Key Findings

- Coupling weather + fatigue into P_{comms} is an actionable lever
- **H1/H3 confirmed** on Blind Shore: fewer fatals (≈37%) and collisions (≈12%)
- **H2 confirmed** everywhere: Proposed raises preparedness vs Classical
- Storm Corridor is harsh on all methods — shared environmental blackout

Limitations & Next Steps

Limits

- Homogeneous crew $n = 20$, scalar F
- Simplified give-way (lower id)
- SIMCOL / Koopman surrogates
- No live RCC asset contention

Future

- Heterogeneous manning / watch bills
- Richer COLREGs roles
- Contested SAR resources
- Transfer beyond Baltic–North Sea

Thank You!

Questions?

`github.com/Michael-Pytel/maritime-abm`

{pawel.pozorski, michal.pytel2}@pw.edu.pl